

SENIOR PROJECT FINAL REPORT



CS 4850 – Spring 2026

SP-33 – Red

Mindful Micro-Journaling App

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Website: <https://shreenap07.github.io/mindful-journaling-website/>

GitHub: <https://github.com/yeshacodes/Mindful-Journaling-App>

Project Link: <https://mindful-journaling-app.vercel.app/>

Demo Link: <https://www.youtube.com/watch?v=aFJ5IMU483s>

STATS AND STATUS

Lines of Code:

Components/Tools: React, Supabase, OpenAI API, GitHub

Hours Estimate: 200

Hours Actual: 150

Project Status: 100% Complete and Working as Designed

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1. Introduction

1.1 Overview

Daily reflection is an important part of maintaining mental well-being. The Mindful Micro-Journaling App helps users reflect on their thoughts and emotions with short, simple journal entries. The app encourages regular journaling by providing daily writing prompts and an easy way to track moods.

The application combines journaling, mood tracking, and data visualization to help users identify emotional patterns over time. Optional voice-to-text functionality allows users to quickly capture thoughts without typing. To ensure privacy, all journal entries are securely stored using encryption.

The goal of this project is to provide a simple, accessible, and secure platform that promotes mindfulness and everyday mental wellness.

1.2 Project Goals

The primary goal of this project is to create a user-friendly journaling application that encourages consistent reflection. The system is designed to support mood tracking over time, provide optional AI-generated prompts, and ensure strong privacy protections. Additionally, the application focuses on maintaining a clean and minimal interface while supporting both desktop and mobile browsers.

1.3 Definitions

- **Journal Entry:** A short reflection written by the user
 - **Prompt:** A guided question for journaling
 - **Mood Tracking:** Recording emotional states
 - **API:** Interface for external services
 - **Supabase:** Backend service for authentication and storage
-

2. Requirements

2.1 Functional Requirements

The system includes user authentication, journaling features, mood tracking, AI prompt generation, and search/filter functionality. Users can create, edit, delete, and view journal entries, while also tracking moods and analyzing trends.

2.2 Non-Functional Requirements

The application ensures security through authentication and HTTPS communication. It is designed for usability with a simple interface and fast performance. The system supports multiple users and ensures reliability by preventing data loss.

3. Analysis & Design

3.1 System Architecture

The system follows a client-server architecture:

[User]



[React Frontend]

↓ API Calls

[Supabase Backend]



[Database + Authentication]



[Optional OpenAI API]

3.2 Subsystems

- Journal Entry Module
- Mood Tracker Module
- Prompt Generator
- Data Visualizer
- Security Manager
- Notification Manager

3.3 UI Flow Diagram

Login → Home → Journal Entry → Mood Selection → Save
↓
View Trends → Charts

4. Development

The application was developed using React for the frontend and Supabase for backend services. The frontend handles user interactions such as logging in, writing entries, and viewing trends. The backend manages authentication and securely stores user data.

Each component was built modularly to ensure scalability and maintainability. The journaling feature was implemented using React forms, while mood tracking was integrated into each entry. API calls were used to connect the frontend and backend.

Optional AI functionality was implemented using external API calls to generate journaling prompts. This feature is only triggered when requested by the user.

Challenges

- Ensuring secure data storage
- Preventing data loss during submission
- Handling API delays

Solutions

- Input validation
 - Secure API handling
 - Retry mechanisms for AI prompts
-

5. Testing (Plan & Report)

5.1 Test Plan

Testing included functional, integration, and regression testing. Core features such as login, journaling, mood tracking, and data storage were validated.

5.2 Test Coverage

- Authentication testing
- Journal entry creation and retrieval
- Mood tracking accuracy
- API connectivity

5.3 Results

All major features were tested successfully. The system performed reliably under normal usage conditions.

5.4 Scope

The following features and functionalities are included in the scope of testing for the Mindful Micro Journaling App:

- User authentication (account creation, login, logout, password recovery)
- Journal entry creation, editing, saving, and deletion
- Mood tracking functionality and storing mood data with entries
- Viewing past journal entries in chronological order
- Search and filter features for entries
- Basic user interface functionality and navigation between pages
- Data storage and retrieval using the backend

- Security checks to make sure users can only access their data
- Optional AI prompt generation feature

Testing will have functional testing, integration testing, and basic user acceptance testing to make sure the system behaves correctly during normal usage.

5.5 Out of scope

The following items are not included in the scope of testing for this product:

- Advanced AI prompt accuracy or quality evaluation
- Performance testing under heavy load
- Mobile app-specific testing
- Detailed security penetration testing only (only basic security validation is included)
- Third-party service reliability
- Full usability studies with real users

These areas are excluded because of project scope, time limitations, and the focus on core functionality.

5.6 Analysis of Scope and Test Focus Areas

Release Content

The Mindful Micro Journaling App has the core features developed in the prototype. These features focus on providing a simple and effective journaling experience for users.

The main content of this release includes:

- User authentication
- Journal entry creation and saving
- Mood tracking functionality linked to entries
- Viewing past journal entries
- Basic user interface and navigation
- Backend integration using Supabase for data storage
- Partial implementation of the AI prompt feature

This release focuses on validating the core functionality of the application and making sure that users can successfully perform basic journaling tasks.

5.7 Regression Testing

Regression testing will be performed to make sure that existing features are continuing to work correctly after any updates or changes are made to the system.

Because the application is being developed incrementally, regression testing is important to verify that adding new features like mood tracking or AI prompts do not break the previous working features like login or journal entry creation.

Key areas for regression testing include:

- Login and authentication functionality
- Saving and retrieving journal entries
- Mood tracking data storage
- Navigation between pages

Regression testing will be performed after updates to make sure that all previously working features are stable and functional.

5.8 Platform Testing

Testing for this project needs to be performed on a web-based platform, because the application is currently implemented as a web application.

The platform used for testing includes:

- Operating System: Windows or macOS
- Browser: Google Chrome, with basic testing on other browsers if available
- Hardware: Standard laptop or desktop computer
- Internet connection required for backend services and AI features

The application will be tested to make sure it functions correctly within a web browser environment and that all features are working as expected under normal usage conditions.

5.9 Progression Test Objectives

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
Function to be tested					
1	User Registration/Login	Verify that users can create an account and log in securely	Successful signup with valid credentials; login works; error shown for invalid login.	FR-1	High
2	Daily Journaling Entry	Ensure users can create and save journal entries	Entry can be written, saved, and retrieved; no data loss after refresh	FR-2 (Journal System)	High
3	Guided Prompts	Verify that daily prompts are displayed and usable	Prompt appears daily; user can respond and save; prompt updates correctly	FR-3 (Prompt Feature)	Medium
4	Mood Tracking	Ensure users can log and view mood data	Mood selection works; data saved; history displayed correctly	FR-4 (Mood Module)	High
5	Mindfulness Exercises	Verify access and completion of mindfulness sessions	Session starts correctly; timer/audio works; completion is recorded	FR-5 (Mindfulness Module)	Medium
6	Notifications	Ensure reminders are triggered properly	Notifications appear at scheduled time; user can enable/disable	FR-6 (Notification System)	Medium

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
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Regression testing					
R1	User Registration / Login	Ensure updates do not break authentication functionality	Users can still sign up/login; no new login errors introduced	Previous Test Suite Ref 1, FR-1	High
R2	Daily Journaling Entry	Verify journaling functionality remains intact after updates	Entries can be created, saved, and retrieved without data loss	Previous Test Suite Ref 2, FR-2	High
R3	Guided Prompts	Ensure prompt system continues to function correctly	Prompts display correctly and update daily; responses save properly	Previous Test Suite Ref 3, FR-3	Medium
R4	Mood Tracking	Confirm mood logging and history tracking still work	Mood entries are saved and displayed accurately over time	Previous Test Suite Ref 4, FR-4	High
R5	Mindfulness Exercises	Ensure mindfulness sessions are not impacted by changes	Sessions start, run, and complete without errors	Previous Test Suite Ref 5, FR-5	Medium

5.9.1 Security

Security testing will be performed to make sure that user data is protected and only authorized users can access their information. This includes verifying that users need to log in to access the application and that each user can only view their own journal entries.

Testing will also confirm that authentication through Supabase works correctly and that sensitive data like passwords is handled securely. Basic checks will be performed by the project team to make sure that there is proper access control and data protection.

5.9.2 Stress & Volume Testing (S&V)

Stress and volume testing will be limited for this project because of scope and time constraints. However, basic testing will be performed to make sure that the application can handle normal usage like multiple journal entries being created and stored.

The expected outcome is that the system can handle typical user activity without crashing or slowing down significantly.

5.9.3 Connectivity Testing (CT)

Connectivity testing will be performed to make sure that the application properly communicates with the backend and external AI API.

This includes verifying that:

- Data is successfully sent and retrieved from the database
- Journal entries are saved and loaded correctly
- AI prompts are generated when requested

Testing will also check how the system behaves if there is a network issue

5.9.4 Disaster Recovery/Back Up

Basic testing will be performed to make sure that data is not lost during normal usage, This includes verifying that journal entries remain saved after logging out and logging back in.

Since the application relies on backend storage, backup and recovery are primarily handled by the backend service. The expected outcome is that user data is available and consistent.

5.9.5 Unit Testing

Unit testing will be performed on individual components of the application to make sure each part works correctly on its own.

Examples are:

- Testing journal entry creation functions
- Testing mood selection functionality
- Testing login validation logic

Unit testing will be performed by developers during the development process.

5.9.6 Integration Testing

Integration testing will be performed to make sure that different parts of the system work together correctly.

This includes:

- Frontend communicating with the backend
- Authentication working with data storage
- AI prompt feature integrating with the user interface

Test Level	External Party	Proj Team	Business
Unit Testing		P	
Integration Testing		P	
Security Testing	P	S	
Connectivity Testing		P	
User Acceptance Testing		S	P
Production Verification Testing		S	P

5.9.7 Test Type & Approach

Test Type	Objectives
Progression Requirements	The objectives are to verify that the application: • Meets the defined requirements; • Performs and functions accurately; • Correctly handles error conditions; • Interfaces function correctly; • Data load is successful. Functional testing will occur in an iterative and controlled manner, ensuring the solution matches the defined requirements.
Regression testing	The objective of regression testing is to make sure that previously developed and tested features continue to work correctly after system updates or changes. This includes verifying that core functionalities like login, journal entry creation, mood tracking, and navigation are not affected by new features or bug fixes. Regression testing will be performed after updates to confirm that no new issues have been introduced and that existing features remain stable.

5.9.8 Build strategy

The system will be implemented using an incremental build strategy, where core functionality is developed and tested first, followed by additional features in stages. The initial build focuses on essential components such as user authentication, journaling, functionality, and data storage. Subsequent builds introduce advanced features including guided prompts, mood tracking, mindfulness exercises, and notifications. This phased approach allows for early testing and validation of critical system components while reducing integration risks. Each functional module is tested independently before being integrated into the overall system, to make sure that it is stable and is well maintained throughout the development.

Test Execution Schedule

The testing process will follow a structured sequence to ensure all components are validated effectively.

- Unit Testing (High Priority) : Conducted during development to verify individual functions such as login, journaling, and mood tracking.
- Integration Testing (High Priority) : Perforated after unit testing to ensure modules work together correctly.
- System Testing (Medium Priority) : Validates the entire application workflow, including navigation, data flow, and feature interaction.

Facility, data, and resource provision plan

The test environment for the Mindful Micro Journalling app will be a web-based setup. Testing will be performed using a standard laptop or desktop with access to the internet.

The application will run in a web browser and connect to the backend services for authentication and data storage. The test environment will be available throughout the development and testing phase to allow continuous testing of the features.

Access to other applications

The following external systems and applications are required during testing:

Supabase , web browser, and an optional AI API for prompt generation.

These systems need to be accessible to make sure that the application can be fully tested, especially for features that depend on backend services and API calls.

Testing Requirements

- A computer with a supported operating system (macOS or Windows)
- Access to the application through a local development environment or deployed test version
- Access to the data system for verifying stored data
- A database management tool
- Access to development tools such as VS Code
- Internet access (Microsoft Excel, or shared document)

Data Requirements

Testing requires sample data to verify system functionality. This includes:

- Test user accounts
- Sample journal entires
- Sample mood selections
- Test inputs for search and filter features

This data will be used to confirm that the system correctly stores, retrieves, and displays information.

Process	Tool
Test case creation	Microsoft Word
Test case tracking	Microsoft Excel
Test case execution	Manual testing(Web Browser)
Test case management	Microsoft Excel
Defect management	Microsoft Excel
Development testing	Visual Studio Code
Version control	Github
Backend testing	Supabase Dashboard
Browser testing	Google Chrome

The testing handover procedure defines how the application is passed between different stages of testing during development.

After developers complete initial development and unit testing of a feature, the

system will be handed over to the testing phase for further validation. The testing team will then perform functional and integration testing to make sure that the feature works correctly.

If defects are found during testing, they will be documented and reported back to the developers. The developers will fix the issues, and the updated version will be re-tested to make sure the problem has been resolved.

Once all major features have been tested and no critical issues are there, the application is considered ready for final review and submission. This process makes sure that all components are properly tested and they meet the project requirements before completion.

5.9.9 Testing Metrics

To evaluate the effectiveness of the testing process and the overall quality of the application, several key metrics will be captured throughout the testing phase. One of the main metrics is the total number of defects raised, which helps measure the overall system quality and highlights how stable the application is. In addition, defects will be tracked by module, journaling, mood tracking, and mindfulness to identify which parts of the system are most error-prone and may require further improvement.

6. Version Control

GitHub was used for version control. The team followed a collaborative workflow where each member contributed through commits and pull requests. Changes were tracked and merged systematically to maintain stability.

7. Summary

The Mindful Micro-Journaling App successfully meets its objectives by providing a simple, secure, and effective platform for daily reflection. The integration of journaling, mood tracking, and optional AI features creates a well-rounded user experience.

Future improvements may include enhanced visualizations, personalization features, and improved performance.

8. Appendix

A. Project Plan

- Weekly meetings (Tuesdays & Thursdays)
- GitHub for task tracking

B. Tools Used

- React
- Supabase
- OpenAI API
- GitHub

C. Installation Steps

1. Clone repository
2. Run npm install
3. Add environment variables
4. Run npm run dev

D. Diagrams

[illegible]